

Ego-EXTRA: video-language Egocentric Dataset for EXpert-TRAinee assistance

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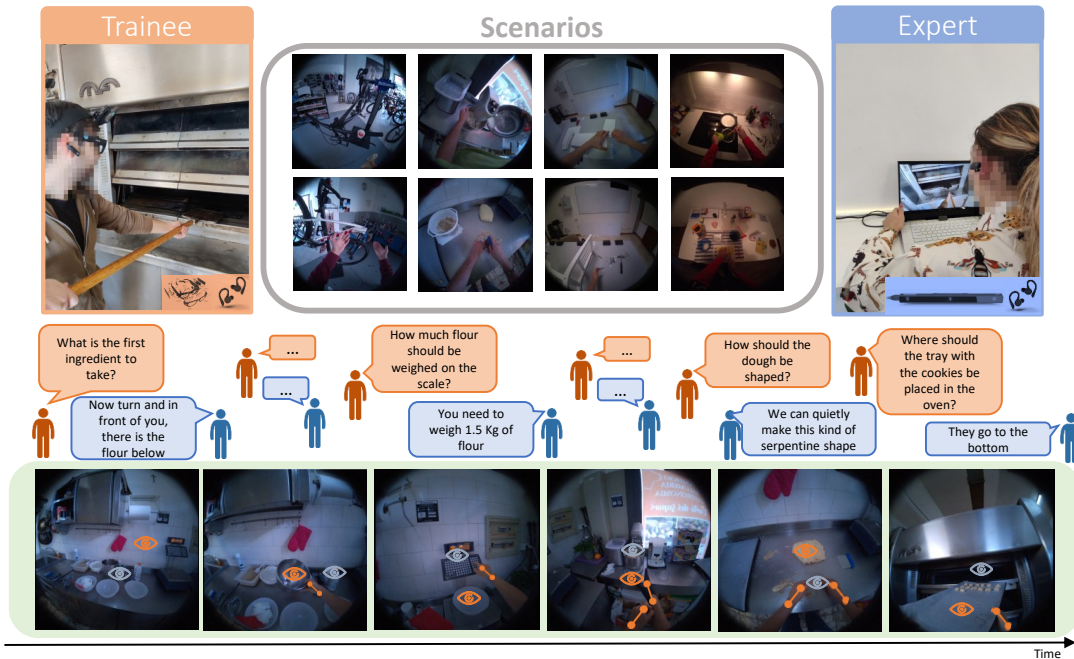


Figure 1. We collect egocentric videos of **trainees** (left) performing procedures while aided by an **expert** (right) enacting a wearable visual assistant which observes the scene from the trainees’ point-of-view and provides guidance through natural language. We gather transcripts of rich natural language dialogue, plus different multimodal signals including eye gaze of both **trainee** (T) and **expert** (E), hand keypoints, SLAM, and IMU. The result is a unique set of videos with temporally-aligned dialogue and multimodal signals gathered by Aria glasses.

Abstract

We present *Ego-EXTRA*, a video-language Egocentric Dataset for EXpert-TRAinee assistance. *Ego-EXTRA* features 50 hours of unscripted egocentric videos of subjects performing procedural activities (the trainees) while guided by real-world experts who provide guidance and answer

specific questions using natural language. Following a “Wizard of OZ” data collection paradigm, the expert enacts a wearable intelligent assistant, looking at the activities performed by the trainee exclusively from their egocentric point of view, answering questions when asked by the trainee, or proactively interacting with suggestions during the procedures. This unique data collection protocol enables *Ego-EXTRA* to capture a high-quality dialogue in which expert-level feedback is provided to the trainee. Two-way dialogues between experts and trainees are recorded,

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transcribed, and used to create a novel benchmark comprising more than 45k high-quality Visual Question Answer sets, which we use to evaluate Multimodal Large Language Models. The results show that Ego-EXTRA is challenging and highlight the limitations of current models when used to provide expert-level assistance to the user. The Ego-EXTRA dataset is publicly available to support the benchmark of egocentric video-language assistants: <https://fpv-iplab.github.io/Ego-EXTRA/>.

1. Introduction

Every day, people naturally engage in different activities, such as washing a car, repairing a bike, assembling new furniture or preparing dinner. Mastering most of these activities requires time, dedication and, often, the guidance of an expert. Think of a college student learning to cook their favorite dish from their father or how to fix a running toilet from their mother. Although the web provides plenty of resources to learn such skills autonomously [2, 5, 6], wearable devices equipped with vision and computation abilities, such as smart glasses, have the potential to act as a *world expert*, providing guidance in a natural way [48]. Thanks to their ability to look at the world from the privileged egocentric point of view of the user, wearable assistants can relate vision to language and contextualize questions like “What is this?” or “What should I do now?”. Furthermore, wearable assistants should be able to provide practical suggestions that guide the user through the steps of repairing a bike chain and checking that each step has been performed safely and correctly. Toward this direction, previous works investigated tasks related to procedural video understanding, such as keystone recognition [29, 56], mistake detection [25, 43, 52], planning [32], procedure understanding [29], and proficiency estimation [21, 29], as well as in tasks related to natural language processing in egocentric vision [12, 14, 42]. Egocentric vision has also seen a significant increase in the availability of large-scale datasets [16, 28, 29, 56]. Although these datasets supported the development of procedural video understanding, they do not capture naturalistic dual-agent conversations paired with realistic egocentric visual observations (see Figure 1). Indeed, they typically include textual information obtained through post-acquisition narrations by the camera wearer [17], third-party annotators [28], or experts’ commentary [29], which makes current video source not directly aligned to the objective of evaluating the performance of wearable procedural assistants.

To address this limitation, we present Ego-EXTRA, a new dataset of EXpert-TRAinee interactions aimed at validating video-language models. Ego-EXTRA is composed of 50 hours of egocentric procedural videos with real trainee-expert conversations recorded during the video ac-

quisition process. The dataset has been collected following the “Wizard of OZ” paradigm historically adopted in experimental psychology [37], linguistics and dialogue state tracking [45], where a human simulates a machine interacting with a user. In our setting, as shown in Figure 1, a *Trainee* wears ARIA glasses [55] to acquire data while performing a given procedural activity, while an *Expert* observes the scene from the trainee’s point of view through a laptop, providing them with assistance and answering questions. We considered four scenarios (i.e., bike workshop, kitchen, bakery, and assembly) where trainees performed different activities (e.g., replacing bike brake pads or cooking a tart), asking questions to the expert whenever they needed help. Thanks to ARIA’s rich sensor suite, we simultaneously captured different signals, including RGB, SLAM, eye gaze, IMU, magnetometer, barometer, GPS, BLE, Wi-fi, hand keypoints, and audio enriching the dataset and aligning to previous data collection protocols [29]. Conversations between trainees and experts are transcribed to text in order to provide language supervision, resulting in a novel set of egocentric videos associated with dual-agent conversations temporally aligned to videos, a significant departure from current acquisition protocols.

Based on the natural conversations included in Ego-EXTRA, we designed a benchmark of 45.198 realistic visually grounded question-answer sets (QA sets) and validate the ability of current Multimodal Large Language Models (MLLMs) in supporting the user with natural language supervision. To build the benchmark we designed a novel protocol based on the extraction, automatic generation and manual validation/refinement of QA sets which is scalable and applicable to future collection efforts. We evaluated 4 state-of-the-art visual-language models on the proposed VQA benchmark and thoroughly examined their limitations. For comparison, we also evaluated a total of 5 LLMs using only textual input, highlighting that QA sets are based on video content. Results show that MLLMs achieve an average accuracy ranging from 29.21% to 41.38%, demonstrating that the proposed VQA benchmark is challenging for current methods. The dataset will be publicly released to support research in this area.

In sum, the contributions of this work are: 1) we present Ego-EXTRA, a new dataset acquired in realistic scenarios that comprises 50 hours of egocentric videos and naturalistic trainee-expert conversations; 2) we build a challenging VQA benchmark with a rigorous human-validation step to ensure the high-quality of QA sets. The benchmark is designed as a test set to validate the assistive ability of models; 3) we evaluate different LLMs and MLLMs on the proposed benchmark to assess their performance when answering trainee’s questions; 4) we will release the dataset to support the research community in evaluating visual-language models aimed at assisting humans in real-world scenarios.

2. Related Work

Expert-Level Assistance in Egocentric Vision An appealing feature of wearable egocentric systems is their potential to provide expert-level support to human activities [36, 48]. Previous works investigated a plethora of individual tasks aimed to support the development of such systems, notably including temporal action segmentation [53, 54, 68], action anticipation [27, 44, 70], mistake detection [33, 52, 65], procedure understanding [7, 51, 52, 71], proficiency estimation [29], and skill determination from video [21, 22]. While these tasks provide essential building blocks to enable the development of assistive systems, an holistic benchmark to support the development and evaluation of methods is missing. Towards this direction, we propose Ego-EXTRA, the first dataset of egocentric videos centered around natural vision-language dialogue interactions between experts and trainees aimed to support the evaluation of systems for user assistant in procedural tasks.

Language-Based Egocentric Vision Datasets Egocentric vision datasets have often included forms of natural language supervision, usually collected after video acquisition [17, 28, 29]. Datasets of natural conversations of human-object interactions have also been proposed [46]. Other works included natural language data in the form of procedural instructions [34, 47, 50]. While providing natural language data at various levels, these previous works did not explicitly aim to collect the specific language of the expert in natural conversations with the user. Notably, HoloAssist [65] and Ego-Exo4D [29] recently proposed data collection paradigms aimed at including instructor or expert language respectively. In particular, HoloAssist [65] includes videos of trainees following a given procedure, supported by an instructor who gives them guidance in natural language. The natural language data is used to seed labels for a number of tasks, including action recognition, mistake detection, and intervention prediction, but the raw natural language data is not publicly available. Ego-Exo4D [29] includes videos of subjects with different levels of expertise performing given procedures autonomously. The experts narrate videos after the acquisition, highlighting areas of improvements and good task executions. Similarly to HoloAssist, we collect natural conversations between a trainee and a supervisor. While HoloAssist focuses on simple procedures, we target real-world procedures such as repairing a bike and making a tart, and recruit real-world experts similar to Ego-Exo4D. Differently from Ego-Exo4D, we aim to collect real dialogue between experts and trainees *during the execution of the task*, with the aim of capturing all nuances of human-assistant dialogue and provide a realistic benchmark for assistive egocentric vision systems. Table 1 compares Ego-EXTRA (bottom row) with other existing state-of-the-art datasets (top rows).

Egocentric Vision-Language Benchmarks Owing to the

surge in popularity of language models [23, 61–63], several works proposed benchmarks of egocentric videos based on visual question answering. Typical paradigms for generating high-quality Visual Question Answer (VQA) samples for model evaluation are leveraging synthetic data [26, 57, 66], and augmenting existing datasets of real egocentric videos with human annotations [66], either with automatic or semi-automatic generation [10, 11, 13, 14, 31, 35, 42, 67]. We follow a similar paradigm to generate a curated VQA dataset from the natural language conversations of Ego-EXTRA with the goal of providing a scalable benchmark for vision-based assistive systems communicating with users using natural language. Table 1 compares Ego-EXTRA (bottom row) with other existing state-of-the-art VQA benchmarks (middle rows).

Multimodal Large Language Models Recent works have focused on enhancing Large Language Models (LLMs) to build Multimodal LLMs capable of processing both vision and language data to tackle complex tasks such as question-answering [11, 20], open-ended questions [8] and more general tasks [39, 64] like visual question answering, document reading, and mathematical reasoning. Additionally, there is significant interest in evaluating the abilities of MLLMs on zero-shot downstream tasks, without retraining on specific target data [15, 40, 58]. We benchmark a set of recent MLLMs on Ego-EXTRA. Our results highlight the limited performance of current MLLMs in assisting humans in realistic scenarios.

3. Data Collection

General Setup We collected Ego-EXTRA following the “Wizard of OZ” paradigm [37], historically adopted in the dialogue state tracking literature [45] to collect realistic conversation turns between users and machine-like systems enacted by humans. Each session involves two participants: a **trainee** performing a procedural activity, such as assembling a chair, and an **expert** who provides guidance and answers questions to ensure correct execution. The trainee wears a custom rig (see Figure 1-left) consisting of Aria glasses [55], a smartphone positioned to capture a similar viewpoint, and a set of earbuds. The Aria device records multiple signals, including RGB videos, SLAM, eye gaze, IMU, and hand keypoints, while the smartphone and earbuds enable communication with the expert. The expert, located in a separate room, observes the trainee’s actions through a laptop that streams the egocentric video feed and communicates with them via earbuds using natural language. To further capture expert behavior, the laptop is equipped with a Tobii Pro Fusion Bar [60] that records their gaze as they watch the video stream and interact with the trainee. The bidirectional audio conversation is recorded and later synchronized with the collected egocentric videos. Locating trainee and experts in different physical rooms en-

	Name	Settings/Environment	Scenarios	Val&Test Hours	avg. video duration (min)	Expert-Trainee Conversations	Modalities	QA/Instruction
Datasets	EPIC-Kitchens-100 [17]	Cooking / Real	Kitchens	25.30	N/A	X	RGB	X
	CaptainCook4D [47]	Cooking / Real	Kitchens	94.5	15.26	X	RGB, depth	X
	LEMMA [34]	House / Real	Kitchens and Living Rooms	10.8	2.00	X	RGB, depth	X
	Ego4D [28]	Multi Domain / Real	Multiple scenarios	288.70	24.11	X	RGB, Audio, 3D environments, stereo, gaze, IMU, multi-view	X
	Ego-Exo4D [29]	Skilled Activities / Real	Soccer, Basketball, Dance, Boulderling, Music, Cooking, Bike Repair, Health Care	85.10	15.32	X	RGB, 7-channel audio, IMU, eye gaze, SLAM, 3D environment point clouds, multiview	X
	MECCANO [49]	Industrial-like / Lab	Toy Assembly	3.15	20.79	X	RGB, depth, gaze	X
	Assembly-101 [53]	Industrial-like / Lab	Toy Assembly	66.80	7.10	X	RGB, multi-view, 3D hand-pose	X
	ENIGMA-51 [50]	Industrial-like / Lab	Electrical Boards Repairing	10.35	26.28	X	RGB, 3D models	200 real instructions
	EMQA [18]	Indoor Environment / Synthetic	Exploration	N/A	N/A	X	RGB	441 synthetic VQA pairs
	EgoVQA [24]	Office / Lab	Object Manipulation	0.65	7.5	X	RGB	580 human VQA pairs
VQA Benchmarks	HoloAssist [65]	Assistive Tasks / Lab	Object Manipulation	49.80	4.47	X	RGB, depth, head pose, 3D hand pose, eye gaze, audio	X
	MM-Ego [67]*	Multi Domain / Real	Multiple scenarios	2	0.2	X	RGB	7026 synthetic VQA pairs
	EAGLE [10]*	Multi Domain / Real	Multiple scenarios	N/A	N/A	X	RGB	400K synthetic instructions
	ProMQA* [30]	Cooking / Real	Kitchens	25	6.47	X	RGB	401 synthetic VQA pairs
	VidEgoThink* [12]	Multi Domain / Real	Multiple Scenarios	204	2.74	X	RGB	600 synthetic VQA pairs
	EgoPlan-Bench** [11]	Multi Domain / Real	Multiple Scenarios	N/A	N/A	X	RGB	4,939 synthetic VQA
	EgoTaskQA# [35]	House / Real	Kitchens and Living Rooms	N/A	N/A	X	RGB	40000 synthetic VQA
	EnvQA [26]	House / Synthetic	Kitchens, Living Rooms, Bedrooms, Bathrooms	38.77	0.2	X	RGB	85072 synthetic VQA
	ActPlan-1K [57]	House / Synthetic	Kitchens, Living Rooms, Bedrooms, Bathrooms	N/A	N/A	X	RGB	X
	DDB	Ego-EXTRA	Assistive Procedural Tasks / Real	Bike Workshop, Kitchen, Bakery, Assembly	50	22.78	✓	RGB, SLAM, Trainee eye gaze, Expert eye gaze, IMU, magnetometer, barometer, GPS, BLE, Wi-fi, hand keypoints, and audio

Table 1. Comparison of Ego-EXTRA (bottom row) with other egocentric datasets (top rows) and benchmarks (middle rows). * indicates a benchmark based on Ego4D [28], ^ indicates an extension of EPIC-Kitchens [17], ° refers to an extension of CaptainCook4D [47], and # indicates an extension of LEMMA [34].

sure that 1) the expert perceives the activity solely from the egocentric point of view of the trainee, and 2) all communication occurs strictly through natural language.

Session Acquisition Protocols As the first of its kind, Ego-EXTRA aims to capture high-quality interactions between trainees and experts, where questions are related to the procedure at hand, rather than to the location of objects or any other elements peculiar to the environment that an expert unfamiliar with the setting could not answer. To ensure the relevance of collected interactions, we designed two session acquisition protocols, as detailed below.

Pro-Active protocol (PA). We instruct the expert to engage in conversations with the trainee in a pro-active way, speaking freely and intervening whenever needed, suggesting next steps, giving instructions, correcting mistakes, and providing any information which is deemed necessary. A typical intervention of the expert could be: *E: “Firstly, remove the wheel slowly. You should use the wrench that is in the second chest on your left”*. Following this protocol, the trainee gets acquainted with the procedure, the environment, the location of objects or tools, and their functions. This protocol results in dense interactions, with the expert’s commentary being predominant and with several conversation turns related to locations of objects and functional areas. Videos have an average duration of 29.64 minutes, with 82.4% of the words spoken by the expert and an average of 264 conversation turns per video (see Figure 2).

On-Demand protocol (OD). We implement this protocol only after the trainee has become familiar with the environment, either by completing a session with the pro-active protocol or by watching a pro-active session conducted by another trainee. The trainee is hence instructed to carry out the procedure autonomously, interacting with the expert whenever they need guidance, while the expert is instructed to only answer the trainee’s questions and to in-



Figure 2. Left: Breakdown of hours per collection protocol, procedure, and scenario. Right: number of expert/trainee conversation turns. PA: Pro-Active, OD: On-Demand.

tervene only if a mistake or a potentially dangerous action is about to occur. A typical trainee-expert interaction is: *T: “Which of the two wheels should I remove?” - E: “The front wheel”*. With this protocol, the dialogue is less dense, but the trainee’s questions are predominant, resulting in a more balanced word distribution between the trainee and the expert, with 61.39% of the words spoken by the expert and an average of 142 conversation turns per video. The average duration of videos is 23.41 minutes (see Figure 2).

As we are interested in natural interactions, we reduce the number of pro-active sessions to a minimum, roughly resulting in a 1 : 3 ratio between pro-active and on-demand videos. When constructing our VQA benchmark, we manually filter out all irrelevant conversation turns from pro-active videos.

Scenarios, Subjects, and Statistics We acquired a total of 123 videos amounting to 50 hours at 15 fps with a resolution of 1408x1408 pixels. The recorded procedures are split across 10 different activities and 4 scenarios, have an average length of 25.24 minutes, and include a mean of 177.76

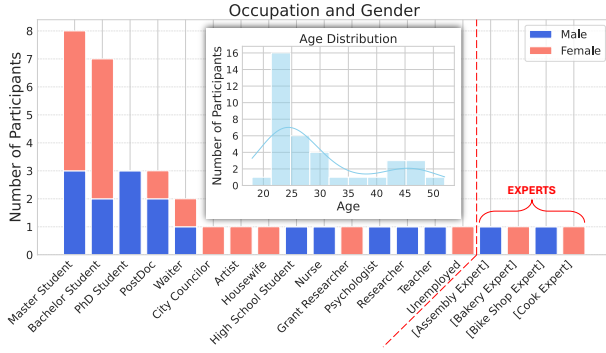


Figure 3. Participants' Demographics.

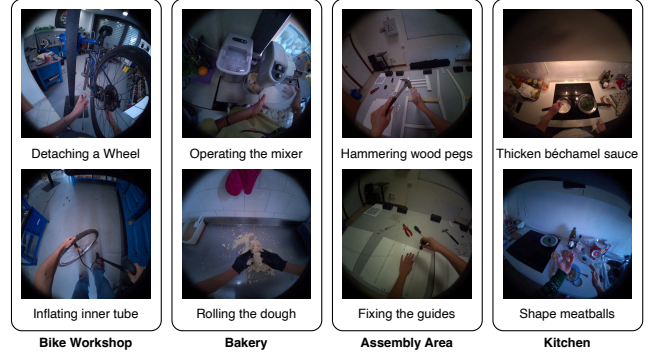


Figure 4. Trainees operate in four scenarios, performing varied activities, interacting with objects and tools, assisted by an expert.



Figure 5. Top verb/noun combinations used by trainees (a) and experts (b). Words reported by category, with examples words in (c).

conversation turns per video, about 422.5 turns per hour or 7 turns per minute. See Figure 2 for a detailed breakdown of video hours and conversation turns per collection protocol, scenario, and activity. Data collection involved a total of 33 trainees and 4 experts across 19 distinct occupations (see Figure 3), all volunteers who provided their privacy consent and authorization to acquire data in the considered environments using the described protocols, transcribe audio conversations, and publicly release the resulting data for research purposes. Experts were selected as professional figures or individuals with consolidated experience in one of the considered procedures, while we chose trainees with no previous experience in the procedures to ensure realistic interactions. While performing the procedures, trainees interact with scenario-specific objects such as an industrial oven in the bakery or inner tubes in the bike workshop, and engage in skilled specialized activities, for which they will require the guidance of the expert (see Figure 4).

Postprocessing Videos collected through Aria devices are sent to Machine Perception Services [1] for the extraction of multimodal signals. We then synchronize Aria videos

with expert's videos and bidirectional audio conversations. The expert's gaze is mapped to Aria's viewpoint¹. All conversations are transcribed using a professional software and translated from the participants' native language to English using Llama 3.1 405B model [23]. The quality of the translations has been manually verified. Each conversation turn has a timestamp which allows to localize it in the video.

Trainee and Expert Language Figure 5 (a-b) summarizes the top-10 verbs and top-5 noun category per verb that trainees and experts use in their conversation turns.² Figure 5 (c) reports a legend for noun categories with example nouns. We note a large use of pronouns ("it", "this", "which"), ambiguous references ("another", "something"), and comparisons ("more", "less", "larger") which suggests that conversations are naturally grounded in video (also see Figure 7). References to quantities, parts, ingredients, tools, and equipment by both trainees and experts denote a precise language and the need for expert guidance. Language is diversified, with trainees mentioning issues more often

¹ See the supplementary material for the details.

² We automatically derive them with NLP techniques.

than experts, while the latter give directions and use task-oriented language.

4. Ego-EXTRA VQA Benchmark

Based on the rich vision-language interactions between experts and trainees present in Ego-EXTRA, we built a VQA benchmark designed to evaluate models following the multiple-choice visual question-answering paradigm [11, 12, 38, 41]. In the following, we use the term *question-answer pair (QA pair)* to denote a question and its correct answer, while the term *question-answer set (QA set)* denotes a set made of a question, the correct answer³ and four wrong answers, which act as distractors. We automatically extract QA sets from transcript. Sets are manually checked and manually refined to ensure quality and grounding in video. Figure 6 illustrates the pipeline we follow for QA set generation and validation, while individual steps are discussed in the following.

Step1: Extraction of Initial Question/Answer Sets While transcripts contain interactions oriented towards a question-answering scheme, automatically mapping conversation turns to QA sets is not trivial due to the use of filler words, informal exchanges, and unstructured conversations. We hence resorted to language models for automated analysis. Specifically, we prompted the Llama 3.1 405B model [23] to 1) extract trainee’s questions and expert’s answers, 2) automatically correct grammar and transcription errors from the original transcripts, and 3) generate an initial set of 4 negative (incorrect) answers for each QA pair. The result is a high-quality set of QA pairs, with initial sets of negatives which will be further refined. While these are generated from textual input, we observe that they are naturally grounded in video thanks to the nature of conversations. Figure 7 reports examples of question-answer pairs automatically extracted from transcripts, while Figure 6 shows an example of initial QA set generated in step 1.

Step 2: Human Validation of QA Sets The initial QA sets were manually reviewed by six human validators to filter out irrelevant, generic, ill-formed, or incorrectly transcribed text. We developed a web-based dashboard to support the manual validation process⁴. Specifically, we show the question, the correct answer, and a set of negative examples. To facilitate annotators during validation, we also included the conversation turn from which the question was extracted, along with the two preceding and two following turns. As shown in Figure 6-Step 2, for each QA set, human validators could indicate if the current QA set is *Acceptable*, *To be Discarded*, *Transcription Error* or *Requires Manual Revision*⁵. Each annotator validated one video for each sce-

nario in Ego-EXTRA. The validation results from this initial set of QAs were then used to design a scalable human validation process on Amazon Mechanical Turk (AMT) to evaluate the entire dataset. Specifically, high-agreement examples from the previous validation phase were used to create a qualification test for AMT workers. Only workers with a global acceptance rate above 90% and a perfect score (100%) on the qualification test were allowed to participate. Each QA pair was validated by five independent workers. With this process, approximately 25% of the initial QAs were discarded.

Step 3: Video Grounding Validation Transcripts of expert–trainee conversations often reference procedural steps and object states that are intrinsically grounded in the video. To ensure this aspect is accurately reflected in the QA sets, we manually reviewed them while also observing the corresponding video clips in which the questions were asked by the trainees (see Figure 6-Step 3). Following the same pipeline adopted for the Step 2, we provided a dashboard to the six annotators to validate the QA sets providing also the video clip associated to the question. For each question, annotators were asked to select one of the following labels: *Grounded*, *Not Grounded*, or *Video Contains the Answer*. As in the previous phase, we extended the validation process to Amazon Mechanical Turk. Only workers with a global acceptance rate above 90% and a perfect score on the qualification test were selected. Each QA pair was validated by five independent workers. Using this pipeline, 28% of the questions were discarded.

Statistics We analyze⁶ the different types of questions and answers in our benchmarks and identify 13 main categories, for which we report statistics and examples in Figure 8. Note that these questions naturally arise from conversations and do not derive from pre-made templates or any bias introduced during prompting. The three most prominent question types are Instructional / Procedural (“What do I do now”), clarifications (“What is the color of the inside?”), and comparisons (“Clockwise or counterclockwise?”). Specific questions about locations (“Does this go here?”), removal (“Can I pull this, right?”), and insertion (“How do I insert the pin), and troubleshooting (“What should I do if the tire is not inflating?”) are also frequent. Questions about confirmations, tool selection, purpose, alignment, suitability, measurement are overall less frequent, but with a large enough minimum number of instances (> 800).

5. Experiments

Baselines We consider four representative Multimodal Large Language Models (MLLMs) as baselines, namely LLaVA-OneVision [39], MiniGPT4-video [8], LLaVa

³Note that each question has only one correct answer.

⁴See the supplementary material for the details.

⁵In this example, annotators flagged the question for revision because all the answer options were written in the first-person singular.

⁶We obtain an initial categorization with language models, then we manually refine.

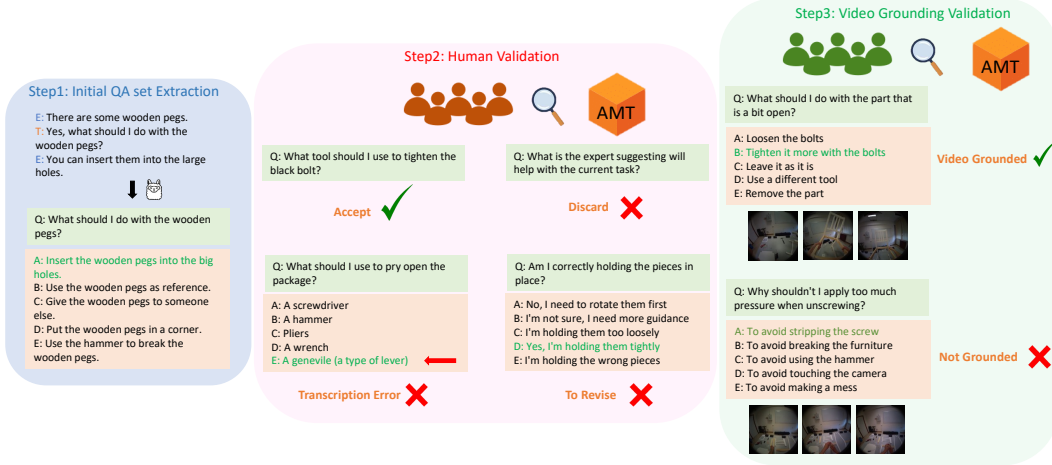


Figure 6. Our multiple-choice question answers generation pipeline: 1) Extraction of QA pairs from transcripts and generation of initial set of negative answers, 2) Human validation, 3) Video Grounding Validation.

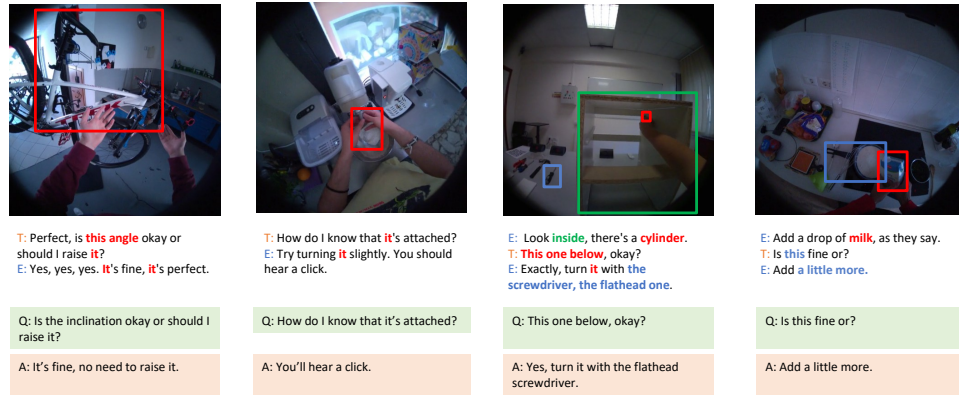


Figure 7. Conversations between trainees and experts are naturally grounded in video since they often refer to objects or their parts. As a result, the generated QA pairs provide a high-quality, visually grounded test-bed for evaluating visual-language models.

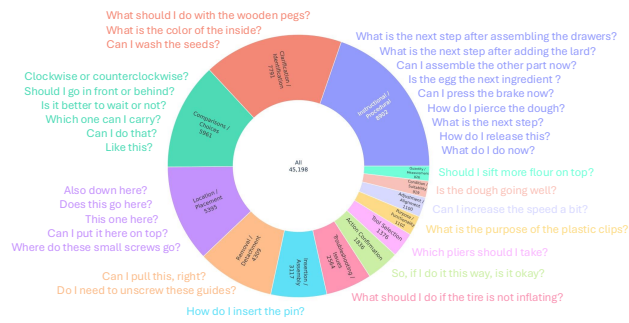


Figure 8. VQA Benchmark Question types and examples.

video [69], and Qwen2.5-VL [9]. We also include five LLMs to assess the ability of current models to infer the correct answer purely from text, without any video context. The considered LLMs are Llama 3.3 Instruct Turbo [4], Llama 3.1 Instruct [3] 8B and 70B, Qwen 2.5 Instruct

72B [59] and DeepSeek-R1 Turbo [19]⁷. All LLMs are prompted with the QA set and asked to output the correct answer. MLLMs also receive a video clip sampled before the question timestamp ($\delta = 5s$). In the main experiments, we feel video models with 8 frames sampled uniformly from the input clip and ablate on other sampling schemes. We additionally provide a human baseline on a sample of 217 questions (an average of 54.25 per scenario)⁸.

Benchmark Results Table 2 reports the overall and per-scenario accuracy of the selected baselines on the VQA benchmark. Language-only models achieve an average accuracy close to a random baseline (20%), ranging from 8.67% with Llama 3.1 Instruct 8B (first row) to 26.65% with Llama 3.1 Instruct 70B (second row). This highlights that Ego-EXTRA is a challenging benchmark for current LLMs. Among Video-Language models, LLaVa-

⁷More details are in the supplementary materials.

⁸see the supplementary material for more details.

Model	Bike Workshop	Bakery	Assembly	Kitchen	Avg.
Llama 3.1 Instruct 8B	07.63	08.62	07.45	10.96	08.67
Llama 3.1 Instruct 70B	27.57	22.54	25.19	31.30	26.65
Llama 3.3 Instruct Turbo	27.14	18.61	24.67	30.42	25.21
Qwen 2.5 Instruct 72B	20.27	15.28	19.01	21.54	19.02
DeepSeek-R1 Turbo	24.22	21.94	21.73	26.15	23.51
MiniGPT4-video	06.62	07.09	08.26	15.74	10.68
LLaVa Video	27.01	27.16	26.12	32.09	28.55
LLaVa-OneVision	32.03	33.13	30.88	35.77	33.06
Qwen 2.5-VL	29.99	28.59	27.47	35.87	31.11
Sample Human Baseline	87.50	90.91	100	81.82	89.65

Table 2. Results on the proposed VQA benchmark. We report the best results in **bold** and the second-best results in underline.

Model	Input	Bike Workshop	Bakery	Assembly	Kitchen	Avg.
MiniGPT4-video	US	06.62	07.09	08.26	15.74	10.68
	QA	06.81 $\uparrow$ 0.19	06.44 $\downarrow$ 0.65	07.95 $\downarrow$ 0.31	16.10 $\uparrow$ 0.36	10.70 $\uparrow$ 0.02
	TS	06.00 $\downarrow$ 0.62	07.42 $\uparrow$ 0.33	08.47 $\uparrow$ 0.21	15.21 $\downarrow$ 0.53	10.45 $\downarrow$ 0.23
LLaVa Video	US	27.01	27.16	26.12	32.09	28.55
	QA	27.62 $\uparrow$ 0.61	27.16 $\pm$ 0.00	26.00 $\downarrow$ 0.12	30.07 $\downarrow$ 2.02	27.89 $\downarrow$ 0.66
	TS	26.02 $\downarrow$ 0.99	23.30 $\downarrow$ 3.86	24.56 $\downarrow$ 1.56	29.37 $\downarrow$ 2.72	26.51 $\downarrow$ 2.04
LLaVa-OneVision	US	32.03	33.13	30.88	35.77	33.06
	QA	26.58 $\downarrow$ 5.45	27.99 $\downarrow$ 5.14	29.41 $\downarrow$ 1.47	34.67 $\downarrow$ 1.10	29.56 $\downarrow$ 3.50
	TS	24.05 $\downarrow$ 7.98	25.57 $\downarrow$ 7.56	21.73 $\downarrow$ 9.15	29.46 $\downarrow$ 6.31	25.29 $\downarrow$ 7.77
Qwen 2.5-VL	US	29.99	28.59	27.47	35.87	31.11
	QA	28.40 $\downarrow$ 1.59	27.23 $\downarrow$ 1.36	25.90 $\downarrow$ 1.57	34.09 $\downarrow$ 1.78	29.48 $\downarrow$ 1.63
	TS	26.58 $\downarrow$ 3.41	26.63 $\downarrow$ 1.96	24.51 $\downarrow$ 2.96	32.99 $\downarrow$ 2.88	28.16 $\downarrow$ 2.95

Table 3. Effect of sampling frames from input video. US: sampling 8 frames uniformly. QA: 8 frames before the timestamp of the question. TS: a single frame at the timestamp. **Increments** and **decrements** computed considering as reference the **US input**.

OneVision and Qwen 2.5-VL obtain similar performance on average (all around $\sim 30\%$). The best performing model is LLaVa-OneVision (third row), which achieves an average accuracy of 33.06%, outperforming other models in each scenario, except for the *Kitchen* scenario, where Qwen 2.5-VL (fourth row) obtains the highest accuracy. Nevertheless, the overall average performance of 33.06% highlights the significant challenge of the proposed VQA benchmark, especially when compared to the human baseline, which achieves an accuracy of 89.65%. See the supp. material for qualitative results.

Importance of Video Sampling Following [72], we analyze three different strategies for sampling frames from the video clip given as input to the visual-language models: *Uniform Sampling* (sampling 8 frames uniformly), *QA frames* (the last 8 frames of the clip), and *TS frame* (a single frame at the question timestamp). This last approach also highlights whether image-level understanding is sufficient to solve our VQA benchmark. Results in Table 3 show that US leads to best results in average, and models generally exhibit performance drops with the QA and TS schemes. Exceptions include the *Bike Workshop* scenario, where MiniGPT4-video and LLaVa Video achieve gains of 0.19% and 0.61%, respectively, with the QA scheme; the *Bakery* and *Assembly* scenarios, where MiniGPT4-video gains 0.33% and 0.21% with TS inputs; and the *Kitchen* scenario, where MiniGPT4-video improves its performance by 0.36% using QA. In general, these results highlight the importance of observing the video clip compared to a sin-

Model	Video Length (s)	Bike Workshop	Bakery	Assembly	Kitchen	Avg.
LLaVa-OneVision	5	32.03	33.13	30.88	35.77	33.06
	15	26.80 $\downarrow$ 5.23	28.59 $\downarrow$ 4.54	25.15 $\downarrow$ 5.73	33.29 $\downarrow$ 2.48	28.69 $\downarrow$ 4.37
	30	26.31 $\downarrow$ 5.72	29.65 $\downarrow$ 3.48	27.52 $\downarrow$ 3.36	34.00 $\downarrow$ 1.77	29.78 $\downarrow$ 3.28

Table 4. Analysis on the effect of input video length uniformly sampled at 8 frames. We selected the best performing model from our VQA benchmark for this analysis. Differences are computed considering as reference the **5 seconds as video length**.

Model	Input	Bike Workshop	Bakery	Assembly	Kitchen	Avg.
LLaVa-OneVision 7B*	QA	17.45	14.98	13.90	19.20	16.62
LLaVa-OneVision 7B*	QA + Transcript	24.71	27.53	33.90	27.53	29.26
LLaVa-OneVision 7B	QA + Video	32.03	33.13	30.88	35.77	33.06
LLaVa-OneVision 7B	QA + Video + Transcript	31.81	33.74	39.78	35.50	36.17

Table 5. Comparison of the best performing LLaVa-OneVision 7B with its language-only counterpart and the addition of the transcript as input. * denotes the language-only counterpart, which takes only text as input.

gle image. To further assess the importance of video length, we evaluate the performance of the best model (LLaVa-OneVision), taking as input 8 frames uniformly sampled from video clips of 5, 15 and 30 seconds. Table 4 shows an average drop (last column) of 4.37% and 3.28% when increasing the temporal spans of video clips to 15 and 30 seconds, respectively. Degradation is observed across all scenarios, where accuracy decreases with longer videos.

Effect of Textual Context Table 5 also compares the best performing model LLaVa-OneVision 7B with its language-only counterpart when feeding the models with the textual transcripts related to the 5-second video clip. This allows to assess the ability of language-only models when they are provided with additional context and simulates an assistant with a basic form of memory of previous conversation turns. The comparison shows how the model benefits from video to answer questions (33.06%) compared to using only textual information (16.62%) or only the transcript (29.26%). Using video and transcript improves average performance, leading to 36.17% accuracy.

6. Conclusion

In this work, we introduced Ego-EXTRA, a novel egocentric dataset designed to validate intelligent wearable assistants that can provide natural language guidance in real-world scenarios. Through realistic expert-trainee conversations, Ego-EXTRA captures the complexities of procedural tasks across various skill-based domains. Based on real conversations, we designed a VQA benchmark and evaluated a range of LLMs and MLLMs, highlighting the challenging nature of the benchmark, the current limitations of text-based LLMs, and the benefits of contextualized video for MLLMs. Data and benchmark are publicly available to support the community in the validation of wearable assistants able to provide language-based expert-level guidance to users.

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References

- [1] Aria mps. https://facebookresearch.github.io/projectaria_tools/docs/ARK/mps. 5
- [2] ehow. <https://www.ehow.com/>. 2
- [3] Llama 3.1 instruct, . <https://huggingface.co/meta-llama/Llama-3.1-8B-Instruct>. 7
- [4] Llama 3.3 instruct turbo, . <https://huggingface.co/meta-llama/Llama-3.3-70B-Instruct>. 7
- [5] wikihow. <https://www.wikihow.com/Main-Page>. 2
- [6] youtube. <https://www.youtube.com/>. 2
- [7] Kumar Ashutosh, Santhosh Kumar Ramakrishnan, Triantafyllos Afouras, and Kristen Grauman. Video-mined task graphs for keystone recognition in instructional videos. *Advances in Neural Information Processing Systems*, 36, 2024. 3
- [8] Kirolos Ataallah, Xiaoqian Shen, Eslam Abdelrahman, Essam Sleiman, Deyao Zhu, Jian Ding, and Mohamed Elhoseiny. Minigt4-video: Advancing multimodal llms for video understanding with interleaved visual-textual tokens, 2024. 3, 6
- [9] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhao-hai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. 7
- [10] Jing Bi, Yunlong Tang, Luchuan Song, Ali Vosoughi, Nguyen Nguyen, and Chenliang Xu. EAGLE: egocentric AGgregated language-video engine. *arXiv*, 2409.17523, 2024. 3, 4
- [11] Yi Chen, Yuying Ge, Yixiao Ge, Mingyu Ding, Bohao Li, Rui Wang, Ruifeng Xu, Ying Shan, and Xihui Liu. EgoPlan-bench: Benchmarking multimodal large language models for human-level planning. *arXiv preprint arXiv:2312.06722*, 2023. 3, 4, 6
- [12] Yi Chen, Yuying Ge, Yixiao Ge, Mingyu Ding, Bohao Li, Rui Wang, Ruifeng Xu, Ying Shan, and Xihui Liu. EgoPlan-Bench: benchmarking multimodal large language models for human-level planning. *arXiv*, 2312.06722, 2024. 2, 4, 6
- [13] Sijie Cheng, Kechen Fang, Yangyang Yu, Sicheng Zhou, Bohao Li, Ye Tian, Tingguang Li, Lei Han, and Yang Liu. VidEgoThink: assessing egocentric video understanding capabilities for embodied AI. *arXiv*, 2410.11623, 2024. 3
- [14] Sijie Cheng, Zhicheng Guo, Jingwen Wu, Kechen Fang, Peng Li, Huaping Liu, and Yang Liu. EgoThink: evaluating first-person perspective thinking capability of vision-language models. In *Computer Vision and Pattern Recognition (CVPR)*, 2024. 2, 3
- [15] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng Huat Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. Instructblip: towards general-purpose vision-language models with instruction tuning. In *Proceedings of the 37th International Conference on Neural Information Processing Systems*, Red Hook, NY, USA, 2024. Curran Associates Inc. 3
- [16] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Sanja Fidler, Antonino Furnari, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, et al. Scaling egocentric vision: The epic-kitchens dataset. In *Proceedings of the European conference on computer vision (ECCV)*, pages 720–736, 2018. 2
- [17] Dima Damen, Hazel Doughty, Giovanni Maria Farinella, Antonino Furnari, Jian Ma, Evangelos Kazakos, Davide Moltisanti, Jonathan Munro, Toby Perrett, Will Price, and Michael Wray. Rescaling egocentric vision: Collection, pipeline and challenges for epic-kitchens-100. *International Journal of Computer Vision (IJCV)*, 2022. 2, 3, 4
- [18] Samyak Datta, Sameer Dharur, Vincent Cartillier, Ruta Desai, Mukul Khanna, Dhruv Batra, and Devi Parikh. Episodic memory question answering. 2022. 4
- [19] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shitong Ma, Peiyi Wang, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jiawei Wang, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Meng Li, Miaojuan Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiusi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shutong Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Shengfeng Ye, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wan-jia Zhao, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang,

- Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanhong Xu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025. 7
- [20] Zhikang Dong, Apoorva Beedu, Jason Sheinkopf, and Irfan Essa. Mamba fusion: Learning actions through questioning. *arXiv*, 2409.11513, 2024. 3
- [21] Hazel Doughty, Dima Damen, and Walterio Mayol-Cuevas. Who’s better? who’s best? pairwise deep ranking for skill determination. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6057–6066, 2018. 2, 3
- [22] Hazel Doughty, Walterio Mayol-Cuevas, and Dima Damen. The pros and cons: Rank-aware temporal attention for skill determination in long videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. 3
- [23] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnston, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Lauren Rantala-Yeary, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kar-
- das, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collet, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gouget, Virginie Do, Vish Vogeti, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyan Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaoqing Ellen Tan, Xinfeng Xie, Xuchao Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpiere Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aaron Grattafiori, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alex Vaughan, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Franco, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Damon Civin, Dana Beaty, Daniel Kreymmer, Daniel Li, Danny Wyatt, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Firat Ozgenel, Francesco Caggioni, Francisco Guzmán, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Govind Thattai, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren,

- Hunter Goldman, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Karthik Prasad, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelen, Keqian Li, Kun Huang, Kunal Chawla, Kushal Lakhota, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Maria Tsimpoukelli, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikolay Pavlovich Laptev, Ning Dong, Ning Zhang, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Rohan Maheswari, Russ Howes, Ruty Rinott, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Kohler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vonimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vitor Albiero, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaofang Wang, Xiaojuan Wu, Xiaolan Wang, Xide Xia, Xilun Wu, Xinbo Gao, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yuchen Hao, Yundi Qian, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, and Zhiwei Zhao. The llama 3 herd of models, 2024. 3, 5, 6
- [24] Chenyou Fan. EgoVqa - an egocentric video question answering benchmark dataset. In *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*, page 4359–4366, Seoul, Korea (South), 2019. IEEE. 4
- [25] Alessandro Flaborea, Guido Maria D’Amely di Melenugno, Leonardo Plini, Luca Scofano, Edoardo De Matteis, Antonino Furnari, Giovanni Maria Farinella, and Fabio Galasso. Prego: Online mistake detection in procedural egocentric videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18483–18492, 2024. 2
- [26] Difei Gao, Ruiping Wang, Ziyi Bai, and Xilin Chen. Env-qa: A video question answering benchmark for comprehensive understanding of dynamic environments. In *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, page 1655–1665, Montreal, QC, Canada, 2021. IEEE. 3, 4
- [27] Rohit Girdhar and Kristen Grauman. Anticipative video transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 13505–13515, 2021. 3
- [28] Kristen Grauman, Andrew Westbury, Eugene Byrne, Zachary Chavis, Antonino Furnari, Rohit Girdhar, Jackson Hamburger, Hao Jiang, Miao Liu, Xingyu Liu, Miguel Martin, Tushar Nagarajan, Ilija Radosavovic, Santhosh Kumar Ramakrishnan, Fiona Ryan, Jayant Sharma, Michael Wray, Mengmeng Xu, Eric Zhongcong Xu, Chen Zhao, Siddhant Bansal, Dhruv Batra, Vincent Cartillier, Sean Crane, Tien Do, Morrie Doulaty, Akshay Erapalli, Christoph Feichtenhofer, Adriano Fragomeni, Qichen Fu, Abraham Gebreselasie, Cristina González, James Hillis, Xuhua Huang, Yifei Huang, Wenqi Jia, Weslie Khoo, Jáchym Kolář, Satwik Kotur, Anurag Kumar, Federico Landini, Chao Li, Yanghao Li, Zhenqiang Li, Karttikeya Mangalam, Raghava Modhugu, Jonathan Munro, Tullie Murrell, Takumi Nishiyasu, Will Price, Paola Ruiz, Merrey Ramazanov, Leda Sari, Kiran Somasundaram, Audrey Southerland, Yusuke Sugano, Ruijie Tao, Minh Vo, Yuchen Wang, Xindi Wu, Takuma Yagi, Ziwei Zhao, Yunyi Zhu, Pablo Arbeláez, David Crandall, Dima Damen, Giovanni Maria Farinella, Christian Fuegen, Bernard Ghanem, Vamsi Krishna Ithapu, C. V. Jawahar, Hanbyul Joo, Kris Kitani, Haizhou Li, Richard Newcombe, Aude Oliva, Hyun Soo Park, James M. Rehg, Yoichi Sato, Jianbo Shi, Mike Zheng Shou, Antonio Torralba, Lorenzo Torresani, Mingfei Yan, and Jitendra Malik. Ego4d: Around the world in 3,000 hours of egocentric video. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18995–19012, 2022. 2, 3, 4
- [29] Kristen Grauman, Andrew Westbury, Lorenzo Torresani, Kris Kitani, Jitendra Malik, Triantafyllos Afouras, Kumar Ashutosh, Vijay Baiyya, Siddhant Bansal, Bikram Boote, Eugene Byrne, Zach Chavis, Joya Chen, Feng Cheng, Fugen Chu, Sean Crane, Avijit Dasgupta, Jing Dong, Maria Escobar, Cristhian Forigua, Abraham Gebreselasie, Sanjay Haresh, Jing Huang, Md Mohaiminul Islam, Suyog Jain, Rawal Khiredkar, Devansh Kukreja, Kevin J Liang, Jia-Wei Liu, Sagnik Majumder, Yongsen Mao, Miguel Martin, Effrosyni Mavroudi, Tushar Nagarajan, Francesco Ragusa, Santhosh Kumar Ramakrishnan, Luigi Seminara, Arjun Somayazulu, Yale Song, Shan Su, Zihui Xue, Edward Zhang, Jinxu Zhang, Angela Castillo, Changan Chen, Xinzhu Fu, Ryosuke Furuta, Cristina Gonzalez, Prince Gupta, Jiabo Hu, Yifei Huang, Yiming Huang, Weslie Khoo, Anush Ku-

- mar, Robert Kuo, Sach Lakhavani, Miao Liu, Mi Luo, Zhengyi Luo, Brigid Meredith, Austin Miller, Oluwatumininu Oguntola, Xiaqing Pan, Penny Peng, Shraman Pramanick, Merey Ramazanova, Fiona Ryan, Wei Shan, Kiran Somasundaram, Chenan Song, Audrey Southerland, Masatoshi Tateno, Huiyu Wang, Yuchen Wang, Takuma Yagi, Mingfei Yan, Xitong Yang, Zecheng Yu, Shengxin Cindy Zha, Chen Zhao, Ziwei Zhao, Zhifan Zhu, Jeff Zhuo, Pablo Arbelaez, Gedas Bertasius, Dima Damen, Jakob Engel, Giovanni Maria Farinella, Antonino Furnari, Bernard Ghanem, Judy Hoffman, C.V. Jawahar, Richard Newcombe, Hyun Soo Park, James M. Rehg, Yoichi Sato, Manolis Savva, Jianbo Shi, Mike Zheng Shou, and Michael Wray. Ego-exo4d: Understanding skilled human activity from first- and third-person perspectives. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 19383–19400, 2024. 2, 3, 4
- [30] Kimihiro Hasegawa, Wiradee Imrattanastrai, Zhi-Qi Cheng, Masaki Asada, Susan Holm, Yuran Wang, Ken Fukuda, and Teruko Mitamura. PromQA: question answering dataset for multimodal procedural activity understanding. *arXiv*, 2410.22211, 2024. 4
- [31] Kimihiro Hasegawa, Wiradee Imrattanastrai, Zhi-Qi Cheng, Masaki Asada, Susan Holm, Yuran Wang, Ken Fukuda, and Teruko Mitamura. Promqa: Question answering dataset for multimodal procedural activity understanding, 2024. 3
- [32] Md Mohaiminul Islam, Tushar Nagarajan, Huiyu Wang, Fu-Jen Chu, Kris Kitani, Gedas Bertasius, and Xitong Yang. Propose, assess, search: Harnessing llms for goal-oriented planning in instructional videos. In *European Conference on Computer Vision*, 2024. 2
- [33] Youngkyoon Jang, Brian Sullivan, Casimir Ludwig, Iain Gilchrist, Dima Damen, and Walterio Mayol-Cuevas. Epicent: An egocentric video dataset for camping tent assembly. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 0–0, 2019. 3
- [34] Baoxiong Jia, Yixin Chen, Siyuan Huang, Yixin Zhu, and Song-Chun Zhu. Lemma: A multi-view dataset for learning multi-agent multi-task activities. page 767–786, Berlin, Heidelberg, 2020. Springer-Verlag. 3, 4
- [35] Baoxiong Jia, Ting Lei, Song-Chun Zhu, and Siyuan Huang. Egotaskqa: Understanding human tasks in egocentric videos. (arXiv:2210.03929), 2022. arXiv:2210.03929 [cs]. 3, 4
- [36] Takeo Kanade and Martial Hebert. First-person vision. *Proceedings of the IEEE*, 100(8):2442–2453, 2012. 3
- [37] John F. (“Jeff”) Kelley. Wizard of oz (woz): a yellow brick journey. *J. Usability Studies*, 13(3):119–124, 2018. 2, 3
- [38] Bohao Li, Yuying Ge, Yixiao Ge, Guangzhi Wang, Rui Wang, Ruimao Zhang, and Ying Shan. Seed-bench: Benchmarking multimodal large language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13299–13308, 2024. 6
- [39] Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. Llava-onevision: Easy visual task transfer, 2024. 3, 6
- [40] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023. 3
- [41] Yuan Liu, Haodong Duan, Yuanhan Zhang, Bo Li, Songyang Zhang, Wangbo Zhao, Yike Yuan, Jiaqi Wang, Conghui He, Ziwei Liu, Kai Chen, and Dahua Lin. Mmbench: Is your multi-modal model an all-around player? In *Computer Vision – ECCV 2024*, pages 216–233, Cham, 2025. Springer Nature Switzerland. 6
- [42] Karttikeya Mangalam, Raiymbek Akshulakov, and Jitendra Malik. Egoschema: A diagnostic benchmark for very long-form video language understanding. *ArXiv*, abs/2308.09126, 2023. 2, 3
- [43] Michele Mazzamuto, Antonino Furnari, Yoichi Sato, and Giovanni Maria Farinella. Gazing into missteps: Leveraging eye-gaze for unsupervised mistake detection in egocentric videos of skilled human activities. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8310–8320, 2025. 2
- [44] Himangi Mittal, Nakul Agarwal, Shao-Yuan Lo, and Kwonjoon Lee. Can’t make an omelette without breaking some eggs: Plausible action anticipation using large video-language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 18580–18590, 2024. 3
- [45] Nikola Mrkšić, Diarmuid Ó Séaghdha, Tsung-Hsien Wen, Blaise Thomson, and Steve Young. Neural belief tracker: Data-driven dialogue state tracking. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1777–1788, Vancouver, Canada, 2017. Association for Computational Linguistics. 2, 3
- [46] Curtis Northcutt, Shengxin Zha, Steven Lovegrove, and Richard Newcombe. Egocom: A multi-person multi-modal egocentric communications dataset. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020. 3
- [47] Rohith Peddi, Shivrat Arya, Bharath Challa, Likhitha Pallapothula, Akshay Vyas, Bhavya Gouripeddi, Jikai Wang, Qifan Zhang, Vasundhara Komaragiri, Eric Ragan, Nicholas Ruozi, Yu Xiang, and Vibhav Gogate. CaptainCook4D: A Dataset for Understanding Errors in Procedural Activities, 2024. 3, 4
- [48] Chiara Plizzari, Gabriele Goletto, Antonino Furnari, Siddhant Bansal, Francesco Ragusa, Giovanni Maria Farinella, Dima Damen, and Tatiana Tommasi. An outlook into the future of egocentric vision. *International Journal of Computer Vision*, pages 1–57, 2024. 2, 3
- [49] Francesco Ragusa, Antonino Furnari, and Giovanni Maria Farinella. Meccano: A multimodal egocentric dataset for humans behavior understanding in the industrial-like domain. *Computer Vision and Image Understanding (CVIU)*, 2023. 4
- [50] Francesco Ragusa, Rosario Leonardi, Michele Mazzamuto, Claudia Bonanno, Rosario Scavo, Antonino Furnari, and Giovanni Maria Farinella. Enigma-51: Towards a fine-grained understanding of human-object interactions in industrial scenarios. *IEEE Winter Conference on Application of Computer Vision (WACV)*, 2024. 3, 4
- [51] Tim J. Schoonbeek, Tim Houben, Hans Onvlee, Peter H.N. de With, and Fons van der Sommen. Industreal: A dataset

- for procedure step recognition handling execution errors in egocentric videos in an industrial-like setting. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 4365–4374, 2024. 3
- [52] Luigi Seminara, Giovanni Maria Farinella, and Antonino Furnari. Differentiable task graph learning: Procedural activity representation and online mistake detection from egocentric videos, 2024. 2, 3
- [53] F. Sener, D. Chatterjee, D. Shelepov, K. He, D. Singhania, R. Wang, and A. Yao. Assembly101: A large-scale multi-view video dataset for understanding procedural activities. *CVPR* 2022. 3, 4
- [54] Fadime Sener, Dipika Singhania, and Angela Yao. Temporal aggregate representations for long-range video understanding. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XVI 16*, pages 154–171. Springer, 2020. 3
- [55] Kiran K. Somasundaram, Jing Dong, Huixuan Tang, Julian Straub, Mingfei Yan, Michael Goesele, Jakob J. Engel, Renzo De Nardi, and Richard A. Newcombe. Project aria: A new tool for egocentric multi-modal ai research. *ArXiv*, abs/2308.13561, 2023. 2, 3
- [56] Yale Song, Eugene Byrne, Tushar Nagarajan, Huiyu Wang, Miguel Martin, and Lorenzo Torresani. Ego4d goal-step: Toward hierarchical understanding of procedural activities. In *Advances in Neural Information Processing Systems*, pages 38863–38886. Curran Associates, Inc., 2023. 2
- [57] Ying Su, Zhan Ling, Haochen Shi, Jiayang Cheng, Yauwai Yim, and Yangqiu Song. ActPlan-1K: benchmarking the procedural planning ability of visual language models in household activities. *arXiv*, 2410.03907, 2024. 3, 4
- [58] Chameleon Team. Chameleon: Mixed-modal early-fusion foundation models. *arXiv preprint arXiv:2405.09818*, 2024. 3
- [59] Qwen Team. Qwen2.5: A party of foundation models, 2024. 7
- [60] Tobii. Tobii pro fusion bar. <https://www.tobii.com/products/eye-trackers/screen-based/tobii-pro-fusion>. 3
- [61] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971*, 2023. 3
- [62] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models, 2023.
- [63] A Vaswani. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017. 3
- [64] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge, Y. Fan, K. Dang, M. Du, X. Ren, R. Men, D. Liu, C. Zhou, J. Zhou, and J. Lin. Qwen2-VL: enhancing vision-language models perception of the world at any resolution. *arXiv*, 2409.12191, 2024. 3
- [65] Xin Wang, Taein Kwon, Mahdi Rad, Bowen Pan, Ishani Chakraborty, Sean Andrist, Dan Bohus, Ashley Feniello, Bugra Tekin, Felipe Vieira Frujeri, Neel Joshi, and Marc Pollefeys. Holoassist: an egocentric human interaction dataset for interactive ai assistants in the real world. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 20270–20281, 2023. 3, 4
- [66] Benita Wong, Joya Chen, You Wu, Stan Weixian Lei, Dongxing Mao, Difei Gao, and Mike Zheng Shou. Assistq: Affordance-centric question-driven task completion for egocentric assistant. (arXiv:2203.04203), 2022. arXiv:2203.04203 [cs]. 3
- [67] H. Ye, H. Zhang, E. Daxberger, L. Chen, Z. Lin, Y. Li, B. Zhang, H. You, D. Xu, Z. Gan, J. Lu, and Y. Yang. MM-Ego: towards building egocentric multimodal LLMs. *arXiv*, 2410.07177, 2024. 3, 4
- [68] Chen-Lin Zhang, Jianxin Wu, and Yin Li. Actionformer: Localizing moments of actions with transformers. In *European Conference on Computer Vision*. Springer, 2022. 3
- [69] Yuanhan Zhang, Jinming Wu, Wei Li, Bo Li, Zejun Ma, Ziwei Liu, and Chunyuan Li. Video instruction tuning with synthetic data, 2024. 7
- [70] Qi Zhao, Shijie Wang, Ce Zhang, Changcheng Fu, Minh Quan Do, Nakul Agarwal, Kwonjoon Lee, and Chen Sun. Antgpt: Can large language models help long-term action anticipation from videos? *arXiv preprint arXiv:2307.16368*, 2023. 3
- [71] Honglu Zhou, Roberto Martín-Martín, Mubbasir Kapadia, Silvio Savarese, and Juan Carlos Nieves. Procedure-aware pretraining for instructional video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10727–10738, 2023. 3
- [72] Sheng Zhou, Junbin Xiao, Qingyun Li, Yicong Li, Xun Yang, Dan Guo, Meng Wang, Tat-Seng Chua, and Angela Yao. Egotextvqa: Towards egocentric scene-text aware video question answering, 2025. 8